Summary in Graph

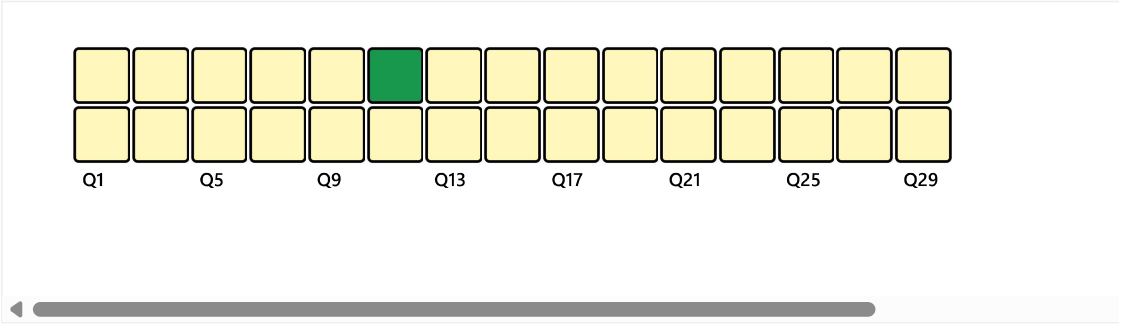
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Subject Wise Test 2)

Qs. Attempted:	1 (0 + 1)	Correct Marks:	2 (0 + 2)
Correct Attempts:	1 (0 + 1)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	2 (0 + 2)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	90 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



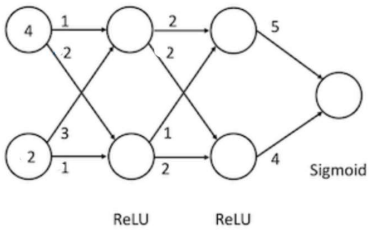
* Average Stats based on first-attempt stats of Top 10% (7) out of 66 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

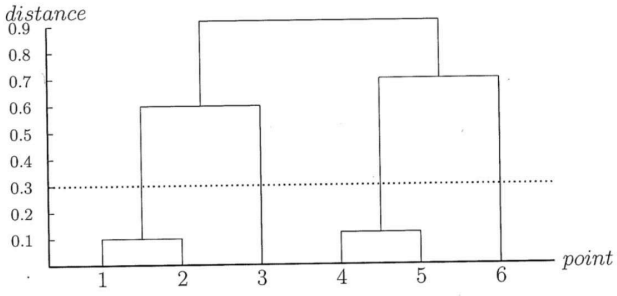
What is the output probability prediction of the following neural network given the input vector shown (rounded to 2 decimal places)?
Both hidden layers have ReLU activation applied to every node . The output layer uses sigmoid activation (for binary classification).



Your Answer: | Correct Answer: 1.0 | Not Attempted | Time taken: 17min 27sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the following dendrogram. How many clusters would be reported if were to slice at the dotted line?



Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following is true about k-means clustering ?

A. k-means diverges and is non-convex.

B. k-means diverges and is convex.

C. k-means converges and is non-convex.

D. k-means converges and is convex.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

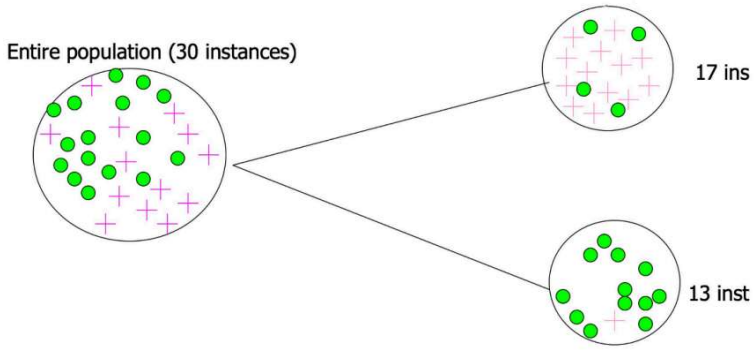
Calculate the misclassification error rate of a decision tree node containing the following dataset:

Class	Count
Class X	40
Class Y	30
Class Z	30

Your Answer: | Correct Answer: 0.6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Find out the weighted average of entropy of children.



Your Answer: | Correct Answer: 0.615 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Let X be a matrix containing data samples as columns, and suppose X is centered such that each feature has a mean of zero. Which of the statements are TRUE?

A. The principal components are eigenvectors of the centered data matrix.

B. The principal components are left singular vectors of the centered data matrix.

C. The principal components are eigenvectors of the sample covariance matrix.

D. The principal components are left singular vectors of the sample covariance matrix.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Select the correct statements about principal component analysis (PCA).

A. PCA is a method of dimensionality reduction

B. If we select only one direction (a one-dimensional subspace) to represent the data, the sample variance of the projected points is zero if and only if the original sample points are all identical

C. The orthogonal projection of a point x onto a unit direction vector w is $\left(x^{\top} w\right) w$

D. If we select only one direction (a one-dimensional subspace) to represent the data, PCA chooses the eigenvector of the sample covariance matrix that corresponds to the least eigenvalue

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Ishani trains two models on the same data with the same features: Model A and Model B. Information on both models is given below:

Model	Bias Squared	Model Variance
A	2	12
B	4	1

Which of the following statements are true? Select all that apply.

A. Model A will overfit more than Model B for this dataset.

B. If both models used LASSO Regression, Model B had the higher λ.

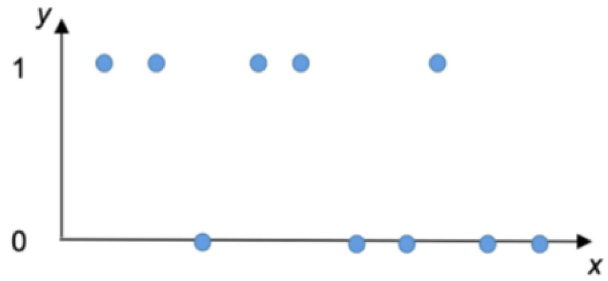
C. Model A could have used Ridge Regression with $\lambda > 0$ while Model B used OLS.

D. Model B must have larger weights than Model A.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

What is the 2-fold Cross-Validation accuracy? Assume the leftmost 5 points (i.e., the 5 points with smallest x values) are in one fold and the rightmost 5 points are in the second fold.



A. 0%

B. 20%

C. 50%

D. 80%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

We are solving a least-squares linear regression problem without regularization. Suppose that the following two weight matrices both have the same cost: $w_1 = \begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix}$ and $w_2 = \begin{bmatrix} -2 \\ 3 \\ 3 \end{bmatrix}$. That is, $\text{RSS}(w_1) = \text{RSS}(w_2)$. Which of the following statements become true if regularization is incorporated into the regression?

A. If l_2 regularization is added with sufficiently high λ, w_1 will be preferred over w_2.

B. If l_∞ regularization is added with sufficiently high λ, w_1 will be preferred over w_2. (Recall that $\|w\|_\infty = \max_i |w_i|$.)

C. If l_1 regularization is added with sufficiently high λ, w_1 will be preferred over w_2.

D. If l_2 regularization is added with $\lambda > 0$, there might be infinitely many weight vectors that minimize $\mathrm{RSS}(w)$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a neural network with the number of neurons in different layers as mentioned in the below list for a regression task. The first layer is the input layer and the last is the output layer.

$$[5, 5, 4, 3, 1]$$

Assuming that there is a bias associated with each neuron, how many total parameters need to be computed?

Your Answer: 73 | Correct Answer: 73 | Correct | Time taken: 02min 27sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following training dataset for a binary classification problem in R^2. Each data-point is represented by $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T$ whose label is y.

Index	x_1	x_2	y
1	1	0	1
2	−1	0	−1
3	5	4	1
4	−5	−4	−1

We wish to train a hard-margin SVM for this problem. $\mathbf{w} = \begin{bmatrix} w_1 & w_2 \end{bmatrix}^T$ represents the weight vector. The index i is for the ith data-point. α_i is the Lagrange multiplier for the ith data-point.

What is the optimal weight vector, $\mathbf{w}^*$?

A. $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

C. $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

D. $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a dataset that has 10 zeros and 5 ones. What is the likelihood function if we assume a Bernoulli distribution with parameter p as the probabilistic model, where $p = P(X = 1)$?

A. p^{15}

B. $(1-p)^{15}$

C. $p^{10} \cdot (1-p)^5$

D. $p^5 \cdot (1-p)^{10}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Assume we are using logistic regression to classify whether a person survived the sinking of titanic or not Given data:

	age	survived	female
0	22.0	0	0
1	38.0	1	1
2	26.0	1	1
3	35.0	1	1
4	35.0	0	0

Suppose after training our model we get

$$\vec{\beta} = \begin{bmatrix} -1.2 \\ -0.005 \\ 2.5 \end{bmatrix}^T$$

where -1.2 is the intercept term, -0.005 is the parameter corresponding to passenger's age, and 2.5 is the parameter corresponding to sex. Consider Silänah Iskandar Näsif Abī Däghir Yazbak, a 20-year-old female. What chance did she have to survive the sinking of the Titanic according to our model? Give your answer as a probability upto 2 decimal places

Your Answer: | Correct Answer: 0.77 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the pairwise distances between 5 points given as matrix. Since the matrix is symmetric, we have given only the lower triangle. Find the final Clustering using single linkage.

	1	2	3	4	5
1	0				
2	2	0			
3	6	3	0		
4	10	9	7	0	
5	9	8	5	4	0

A. $(((1, 2), 3), (4, 5))$

B. $(((1, 2), 5), (3, 4))$

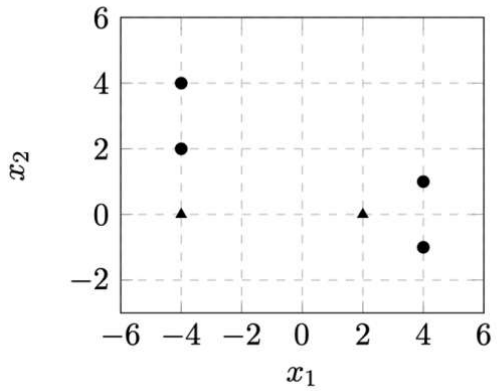
C. $(((1, 2), 4), (3, 5))$

D. $(((2, 3), 1), (4, 5))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Amazing, Inc., runs the k-means clustering algorithm starting from the initial guess shown above, until convergence (no further improvements in loss $L(\mu)$ can be made). Mark the final locations of the stores (with triangles) on the plot below. What is the final loss $L(\mu)$?



Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider this dataset and find out best attribute to split on at the root node based on variance reduction method.

	Ear Shape	Face Shape	Whiskers	Weight
1	Pointy	Round	Present	7.2
2	Floppy	Not Round	Present	8.8
3	Pointy	Not Round	Present	9.2
4	Pointy	Round	Present	8.4
5	Pointy	Round	Absent	7.6
6	Pointy	Round	Absent	10.2
7	Floppy	Round	Absent	15.0
8	Floppy	Round	Absent	11.0
9	Floppy	Round	Absent	18.0
10	Floppy	Round	Absent	20.0

A. Ear Shape

B. Face Shape

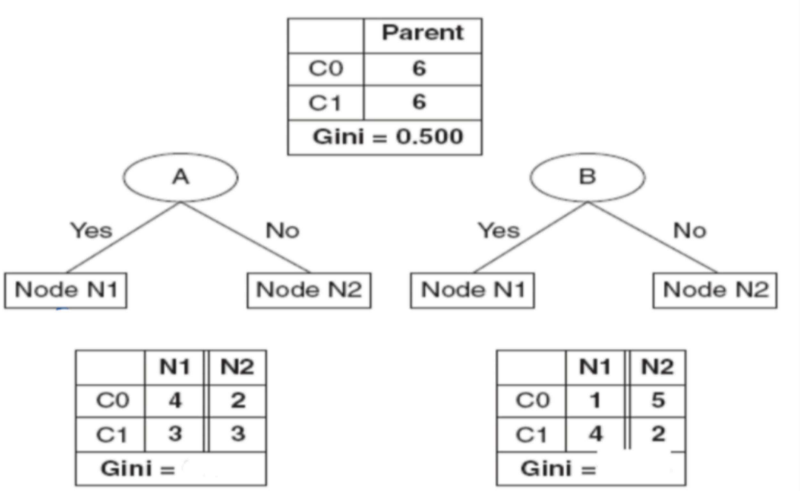
C. Whiskers

D. All attributes are equally likely to split

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

What is the weighted gini average on A and B for the given distribution ?



A. 0.371, 0.486

B. 0.486, 0.371

C. 0.514, 0.629

D. 0.486, 0.629

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

What is the conditional entropy of Y given X_1, i.e., $H(Y \mid X_1)$?

X_1	0	0	1	1
X_2	0	1	0	1
Y	1	1	1	0

A. 0

B. 0.25

C. 0.5

D. 0.75

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

(I) Given a design matrix $X \in \mathrm{R}^{n \times d}$, where $d \ll n$, if we project our data onto a k dimensional subspace using PCA where k equals the rank of X, we recreate a perfect representation of our data with no loss.

(II) PCA produces features that are linear combinations of the original feature.

Select correct option for statement (I) and (II) respectively.

A. (I) is True but (II) is False

B. (I) is False but (II) is True

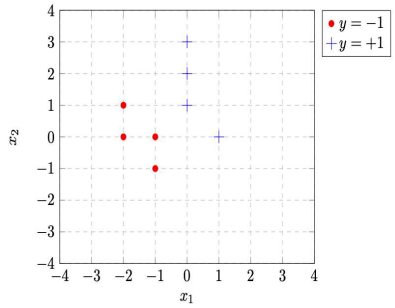
C. Both (I) and (II) are True

D. Both (I) and (II) are False

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Alice plans to apply the Perceptron algorithm t o solve a classification problem on the data set shown in the figure below. For this problem, we will only consider linear separators that pass through the origin.



What is the theoretical upper bound on the number of steps for the Perceptron algorithm to find the linear separator on this data set?

Your Answer: | Correct Answer: 18 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

We will analyze the model bias and variance of a simple linear regression model with no intercept term, $f_\theta(x) = \theta x$. Note that x and θ are both 1-dimensional. Using a simple linear regression model without an intercept (i.e. $f_\theta(x) = \theta x$), we estimate the model variance on a new datapoint $(x, Y) = (3, 12)$ as $\mathrm{Var}\left(f_{\tilde{\theta}}(x)\right) = 3$ using some arbitrary model parameter $\tilde{\theta}$. Suppose we create a new model where $\theta_{\text{new}} = 2\tilde{\theta}$. Calculate the new model variance, $\mathrm{Var}\left(f_{\theta_{\text{new}}}(x)\right)$ using θ_{new}.

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Adrianne also gives Buck some population parameters for the "Score" column in reviews, displayed on the right. Let Y_i denote the "Score" value of entry i in Buck's sample, for $i \in \{1, 2, 3, \ldots, 100\}$.

Assume the underlying data generating process for the "Score" column is a constant model: $Y = \theta + \epsilon$, where $\theta = 7.1$ is a fixed population parameter and ϵ is a random noise with expectation 0 and variance 1.44.
Buck uses the following model to fit his sample of Y_i's: $\hat{Y} = \hat{\theta}$.
Buck estimates θ using the mean of the first 10 "Score" values: $\hat{\theta} = \frac{1}{10}\sum_{i=1}^{10} Y_i$.

Population mean	7.1
Polulation SD	1.2

"Score"

What is his model bias and model variance respectively.

A. 0, 1.44

B. 0, 0.144

C. 7.1, 1.44

D. 7.1, 0.144

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose that we are trying to fit a linear and 10th-degree polynomial to data coming from a cubic function, corrupted by standard Gaussian noise. Let M1 and M2 denote the models corresponding to the linear and 10-degree polynomial. Then,

A. Bias (M1) ≤ Bias(M2), Variance (M1) ≤ Variance(M2).

B. Bias(M1) ≤ Bias(M2), Variance (M1) ≥ Variance (M2).

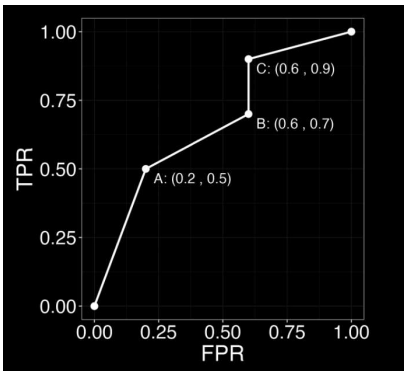
C. Bias(M1) ≥ Bias (M2), Variance (M1) ≤ Variance(M2).

D. Bias(M1) ≥ Bias(M2), Variance (M1) ≥ Variance(M2).

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Fill in the blanks below and explain your answer.



Among the points A, B, and C, _____ is strictly better than _____

A. C, B

B. A, B

C. B, C

D. B, A

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Sara can't believe everyone she knows only decided to watch one of Barbie or Oppenheimer. She collects data and gives each person a label of $y_i = 1$ if a person watched Barbie, and $y_i = 0$ if a person watched Oppenheimer. Assume that everybody in the dataset only watched one of the movies. Suppose Sara believes that wrongly predicting Barbie is worse than wrongly predicting Oppenheimer. Which evaluation metric is the BEST option to maximize in this scenario?

Which evaluation metric is the BEST option to maximize in this scenario?

A. Accuracy

B. Precision

C. True Positive Rate

D. False Positive Rate

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a Support Vector Machine (SVM) with decision boundary $w'x + b = 0$ for a threedimensional feature space, where ' standards for vector transpose. The weight vector is $w = (1 \quad 2 \quad 3)'$, and b=4.

Which of the following points will be classified incorrectly by this SVM ?

A. $x_1 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}', y_1 = -1$ (minus one)

B. $x_2 = (-1 \quad -1 \quad -1)', y_2 = 1$ (plus one)

C. $x_3 = (-3.99 \quad -3 \quad 2)', y_3 = 1$ (plus one)

D. None of the above.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Suppose we have a data set of 100 points whose first few rows are shown below, and that we would like to predict $\vec{y}$ from $\vec{v}$ and $\vec{w}$.Suppose we create a design matrix X whose first column is $\vec{v}$, second column is $\vec{w}$, and third column $\vec{u}$ is a new feature defined by $u_i = |v_i|$. The resulting model is

$$\hat{y}_i = \beta_1 v_i + \beta_2 w_i + \beta_3 |v_i|.$$

The top row corresponds to row 1, for example $y_1 = 4$.

y	v	w
4	-30	1
6	-40	2
5	20	3

For the data above, suppose that

$$\vec{e} \cdot \vec{e} = 9.$$

What is the mean squared error (MSE)?

Your Answer: | Correct Answer: 0.09 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Alice has collected a dataset of dependent and independent variables $\left\{\left(x^{(1)}, y^{(1)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)\right\}$; she does linear regression on it and obtains the weight vector w_{Alice}. Bob also collects the same dataset, but while recording the independent variables (the $x^{(i)}$'s) he uses different units. Specifically, Bob's dataset is $\left\{\left(z^{(1)}, y^{(1)}\right), \ldots, \left(z^{(n)}, y^{(n)}\right)\right\}$, where for each $i, z^{(i)} = cx^{(i)}$, where $c > 0$ is a scalar. Bob does linear regression on this dataset and obtains a weight vector w_{Bob}.

Do Alice and Bob have the same training loss? Is $w_{\text{Alice}} = w_{\text{Bob}}$?

A. Yes, both the training loss and the weights are identical.

B. They may achieve similar training losses, but the learned weights can differ due to variations in initialization or optimization paths.

C. Their training losses differ, yet the final weights turn out to be the same.

D. Both the training loss and the weights are independent, so they can differ arbitrarily.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

First, we use a linear regression method to model this data. To test our linear regressor, we choose at random some data records to be a training set, and choose at random some of the remaining records to be a test set. Now let us increase the training set size gradually. As the training set size increases, what do you expect will happen with the mean training and mean testing errors respectively ?

A. Increase, Decrease

B. Decrease, Increase

C. Increase, Increase

D. Decrease, Decrease

Your Answer: | **Correct Answer: A** | Not Attempted | Time taken: 00min 00sec | Discuss

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